The Irrationality of √2 + √3

Formal status: Kernel verified
Faithfulness: User approved; independently reviewed by claude-opus-5 on claude (agreed)
Informal completeness: Not independently assessed
Document status: TeX compiled

Theorem

The real number $\sqrt{2} + \sqrt{3}$ is irrational; that is, there is no rational number q with $q = \sqrt{2} + \sqrt{3}$.

Formally (Lean 4 / Mathlib):

theorem irrational_sqrt_two_add_sqrt_three : Irrational (Real.sqrt 2 + Real.sqrt 3)

Here ‘Irrational x‘ is Mathlib’s definition ‘x ∉ Set.range ((↑) : ℚ → ℝ)‘, i.e. x is not the image of any rational number under the canonical embedding ℚ → ℝ.

Approved interpretation

- The real numbers ℝ, with its standard complete ordered field structure and the nonnegative square root function Real.sqrt : ℝ → ℝ
- The rational numbers ℚ, embedded into ℝ by the canonical cast Rat.cast : ℚ → ℝ
- Numeric literals 2 and 3 elaborated as elements of ℝ via OfNat
- No user-visible quantifiers: the statement is a closed proposition about one specific real number, so the theorem takes no binders at all.
- The quantification hidden inside ’irrational’ is universal-negative: for every q : ℚ, (q : ℝ) ≠ Real.sqrt 2 + Real.sqrt 3, i.e. the number does not lie in Set.range ((↑) : ℚ → ℝ).
- None; the claim is unconditional and needs no hypotheses beyond Mathlib’s standard theory of ℝ, ℚ, and Real.sqrt.
- Real.sqrt is total in Mathlib (returning 0 on negative inputs); since $2 \geq 0$ and $3 \geq 0$, Real.sqrt 2 and Real.sqrt 3 are the genuine nonnegative square roots.
- Availability of Mathlib’s Irrational predicate (Mathlib.Analysis.SpecialFunctions.Pow / Mathlib.RingTheory.Int.Basic import chain, e.g. via ‘import Mathlib‘).

- Repair of the previous non-elaborating signature: the binders field previously carried the parenthesized prose ’(no binders; closed statement)’, which is not valid Lean binder syntax and would be spliced directly into ‘theorem <name> <binders> : <prop>‘, causing elaboration to fail. The binders field is now the empty string, yielding ‘theorem irrational_sqrt_two_add_sqrt_three : Irrational (Real.sqrt 2 + Real.sqrt 3)‘.

- Fully qualified names are used (Real.sqrt, Irrational) so the statement elaborates without relying on ‘open Real‘; no notation such as √ is used, avoiding dependence on a scoped notation being open.

- ’sqrt(2)’ and ’sqrt(3)’ are the principal (nonnegative) real square roots, formalized as Real.sqrt 2 and Real.sqrt 3 — not two-valued algebraic roots and not elements of an abstract field extension.

- ’is irrational’ is Mathlib’s predicate Irrational x, definitionally $x \notin$ Set.range $((\uparrow) : \mathbb{Q} \to \mathbb{R})$, which is exactly ’not equal to the cast of any rational’.

- ’+’ is addition in $\mathbb{R}$, and the whole sum is the argument of Irrational, so the claim is about the single number Real.sqrt 2 + Real.sqrt 3 — neither weakened (e.g. to irrationality of Real.sqrt 2 alone) nor strengthened (e.g. to linear independence over $\mathbb{Q}$ or a general theorem about sums of square roots).

Human-readable proof

Overview ——– The sum of two irrational numbers need not be irrational (for instance $\sqrt{2} + (-\sqrt{2}) = 0$), so irrationality of $\sqrt{2}$ and of $\sqrt{3}$ separately does not settle the question. The argument below instead assumes rationality of the sum and then *solves algebraically for √2*, reducing the statement to the single classical fact that $\sqrt{2}$ is irrational.

Setup —– Suppose, for contradiction, that $\sqrt{2} + \sqrt{3}$ is rational. By the definition of ‘Irrational‘, this means there exists $q \in \mathbb{Q}$ whose image in $\mathbb{R}$ satisfies

$(q : \mathbb{R}) = \sqrt{2} + \sqrt{3}$. (1)

We use two facts about the real square root, both instances of Mathlib’s ‘Real.sq_sqrt‘ (valid because the radicands are nonnegative):

$(\sqrt{2})^2 = 2$, $(\sqrt{3})^2 = 3$. (2)

Step 1: $q \neq 0$ ————- Since $2 > 0$ and $3 > 0$, both $\sqrt{2} > 0$ and $\sqrt{3} > 0$, hence $\sqrt{2} + \sqrt{3} > 0$. By (1), $(q : \mathbb{R}) > 0$, so in particular $(q : \mathbb{R}) \neq 0$. (In the formal proof this positivity is discharged by the ‘positivity‘ tactic, which knows that $\sqrt{x} > 0$ for $x > 0$.)

Step 2: isolate and square ———————– Rearranging (1) gives

$\sqrt{3} = (q : \mathbb{R}) - \sqrt{2}$.

Squaring both sides and substituting $(\sqrt{3})^2 = 3$ from (2):

$3 = ((q : \mathbb{R}) - \sqrt{2})^2 = q^2 - 2q\cdot\sqrt{2} + (\sqrt{2})^2 = q^2 - 2q\cdot\sqrt{2} + 2,$

where the last equality uses $(\sqrt{2})^2 = 2$. Rearranging the resulting identity yields the key linear relation for $\sqrt{2}$:

$2q\cdot\sqrt{2} = q^2 - 1.$ (3)

Note that this step is purely algebraic: no sign or branch considerations are needed, because we only expand a square and substitute the two identities in (2).

Step 3: solve for √2 and conclude ————————————— By Step 1 we have $q \neq 0$, so $2q \neq 0$ and we may divide (3) by $2q$:

$\sqrt{2} = (q^2 - 1) / (2q).$

The right-hand side is the image in $\mathbb{R}$ of the rational number $(q^2 - 1)/(2q) \in \mathbb{Q}$ — the field operations on $\mathbb{Q}$ commute with the embedding $\mathbb{Q} \to \mathbb{R}$, so the cast of the rational quotient equals the quotient of the casts. Hence $\sqrt{2}$ lies in the range of $\mathbb{Q} \to \mathbb{R}$, i.e. $\sqrt{2}$ is rational.

This contradicts the classical theorem that √2 is irrational (Mathlib's 'irrational_sqrt_two', itself a consequence of the fact that 2 is prime and therefore not a perfect square).

Therefore no rational q with $(q : \mathbb{R}) = \sqrt{2} + \sqrt{3}$ exists, and $\sqrt{2} + \sqrt{3}$ is irrational. ∎

Remarks ——- • A common alternative route squares the sum directly: $(\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6}$, so rationality of the sum would force √6 to be rational, contradicting the irrationality of √6. The route taken here is shorter to formalize because it needs only the irrationality of √2, a named lemma in Mathlib, rather than a fresh non-square argument for 6. • The same technique generalizes: if a and b are distinct nonnegative rationals with √a irrational, the identity above shows √a + √b is irrational, since assuming rationality of the sum lets one solve for √a as a rational expression in the assumed rational value.

Formal status ————- The Lean statement above was rebuilt from the Frozen Claim, checked by a fresh Lean kernel using Mathlib, and its axiom dependencies were audited by Hardy's independent FinalVerifier: the formal proof is kernel verified. The formal script follows exactly the three steps above — positivity of the sum to get $q \neq 0$, squaring to obtain relation (3) (closed by 'nlinarith' from the two square-root identities), then exhibiting $(q^2 - 1)/(2q)$ as a rational witness for √2 and appealing to 'irrational_sqrt_two'.

This prose exposition is a human-readable rendering of that formal argument; the exposition itself has not been independently assessed, and the machine-checked guarantee attaches to the Lean statement and proof, not to the wording here.

Known gaps

- None in the formal development: the Lean proof of the exact Frozen Claim is complete, with no 'sorry', no additional axioms beyond Lean/Mathlib's standard ones, and no auxiliary hypotheses.

- The narrative prose above (including the two Remarks, which are commentary rather than part of the theorem) has not been independently assessed; only the Lean statement and proof were kernel verified.
- The generalization mentioned in the Remarks — that $\sqrt{a} + \sqrt{b}$ is irrational whenever a, b are nonnegative rationals with $\sqrt{a}$ irrational — was not formalized or verified; it is stated as informal commentary only.
- The proof relies on Mathlib's 'irrational_sqrt_two', 'Real.sq_sqrt', and the standard ring/field structure of $\mathbb{Q}$ and $\mathbb{R}$; these are taken as given from the pinned Mathlib revision rather than reproved here.

Exact Lean statement

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theorem irrational_sqrt_two_add_sqrt_three : Irrational (Real.sqrt 2 + Real.sqrt 3)
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Verification appendix

Axioms used: propext, Classical.choice, Quot.sound

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Frozen Claim SHA-256: 371232aaf211dc1fa654c850779771e449297d01f99cc286d6fd1dd320cb1dfc
Lean source SHA-256: 056705a85205dc64acf5084aedad9ceac0fb9f0253a5eaca534d02230eaa473a
Verification SHA-256: 3083bfbb4a5dc6e6d043571a4a9909cae8ce115b8805662d16d63dcede1b143a
Prompt set SHA-256: c418edba223323cd3bbb730ae110d1270be3d23e93994f25a6ddcf2c7f9436a6
Tectonic bundle SHA-256: 6ffe055852f8faf66c0acbe1a7fb27f87b869a90bad1204f3bf4d9683f597c7c
```

Run and tool identities

- Run ID: 8ccb35a8-74bd-4af0-9015-71cf0b1dedca
- Model: claude-opus-5
- Backend: claude
- Runtime SDK: 0.2.127
- Lean: 4.33.1
- Mathlib: 0df444a360eaa60ab8c11dca51a86af692955474
- Tectonic: 0.16.9
- Tectonic bundle: https://data1.fullyjustified.net/tlextras-2022.0r0.tar

Lean kernel verification, faithfulness (approved by the user and read back by an independent model before proof search), heuristic informal review, and document compilation are independent statuses. This trusted-development MVP does not sandbox generated Lean or TeX.